

Harris-Benedict Formula

- For Males:

$$BMR = 66.5 + (13.75 \times \text{weight in KG}) + (5.003 \times \text{height in CM}) - (6.75 \times \text{age})$$

- For Females:

$$BMR = 655.1 + (9.563 \times \text{weight in KG}) + (1.850 \times \text{height in CM}) - (4.676 \times \text{age})$$

(Zajac & Mucha, 2024).

Mifflin-St Jeor Formula

- For Males:

$$BMR = 10 \times \text{weight (in KG)} + 6.25 \times \text{height (in CM)} - 5 \times \text{age} + 5$$

- For Females:

$$BMR = 10 \times \text{weight (in KG)} + 6.25 \times \text{height (in CM)} - 5 \times \text{age} - 161$$

The exercise routine selected is a factor that impacts the BMR calculations by multiplying the result by a figure according to the chosen exercise routine, this is because it accounts for calories burned through daily movement & exercise the multiplications are the same regardless of gender or which formula is used. Refer to the table below for exercise routines & their multipliers.

Exercise Routine Chosen	Multiply BMR Result By(*)
None or very little	*1.2
Light	*1.375
Moderate	*1.55
Heavy	*1.725
Intense	*1.9

BMI Formula

The formula used is the same for males & females.

$$BMI = (\text{Weight in KG} \div \text{Height in CM}) \times 10,000$$

The BMI result is set in different numeric ranges which dictate what weight category an individual is classed as. Refer to the table below for ranges & weight categories.

BMI Range	Weight Category
18 or below	Underweight
18.5 to 24.9	Normal weight (Healthy)
25 to 29.9	Overweight
30 or over	Obese